



ISEC352 – DB Administration and security

sec(351)

Skin care DB

(secure and manage DB)

Group number : 7

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Table of Contents

Introduction:	3
Code:	3
Code Explanation:	
Output:	
References:	

Introduction:

This project is dedicated to securing and managing a database for a medium-sized skin care company, focusing on protecting sensitive employee information. The database will store critical data such as employee numbers, salaries, emails, and birth dates. Through the implementation of access control mechanisms, encryption techniques, and role-based permissions, the project ensures the confidentiality and integrity of this sensitive information. The documentation details the process of database creation, role assignment, data encryption, and security testing to meet best practices in database administration and security.

Code:

```
CREATE DATABASE SkinCare0;
```

```
USE SkinCare0;
```

```
CREATE TABLE Employees (  
    EmployeeID INT PRIMARY KEY,  
    Name VARCHAR(100) NOT NULL,  
    Email VARCHAR(100) NOT NULL UNIQUE,  
    PhoneNumber VARCHAR(15),  
    Position VARCHAR(50) NOT NULL  
);
```

```
CREATE TABLE Salaries (  
    SalaryID INT ,  
    EmployeeID INT NOT NULL,  
    BaseSalary DECIMAL(10, 2) NOT NULL,  
    Bonus DECIMAL(10, 2) DEFAULT 0,  
    PaymentDate DATE NOT NULL,  
    FOREIGN KEY (EmployeeID) REFERENCES Employees(EmployeeID)  
);
```

```
CREATE TABLE Users (  

```

```
UserID INT PRIMARY KEY,  
Username VARCHAR(50) NOT NULL UNIQUE,  
Password VARCHAR(256) NOT NULL,  
Role VARCHAR(20) NOT NULL  
);
```

```
INSERT INTO Employees (EmployeeID, Name, Email, PhoneNumber, Position)  
VALUES  
(60600, 'Sara', 'sara@gmail.com', '0571234506', 'Sales Representative'),  
(60601, 'Layla', 'layla@gmail.com', '0579876556', 'HR Manager'),  
(60602, 'Mona', 'mona@gmail.com', '0573456786', 'Finance Manager'),  
(60603, 'Nada', 'nada@gmail.com', '0571122336', 'Sales Representative'),  
(60604, 'Jana', 'jana@gmail.com', '0575566776', 'Product Manager'),  
(60605, 'Noura', 'noura@gmail.com', '0579988776', 'Marketing Specialist');
```

```
INSERT INTO Salaries (SalaryID, EmployeeID, BaseSalary, Bonus, PaymentDate)  
VALUES  
(1, 60600, 5500.00, 600.00, '2024-11-27'),  
(2, 60601, 8500.00, 1200.00, '2024-11-27'),  
(3, 60602, 9500.00, 2500.00, '2024-11-27'),  
(4, 60603, 5500.00, 350.00, '2024-11-27'),  
(5, 60604, 7500.00, 1700.00, '2024-11-27'),  
(6, 60605, 6500.00, 900.00, '2024-11-27');
```

```
CREATE USER 'regular_user'@'localhost' IDENTIFIED BY 'regular_user_password';  
CREATE USER 'hr_manager'@'localhost' IDENTIFIED BY 'hr_manager_password';  
CREATE USER 'finance_manager'@'localhost' IDENTIFIED BY 'finance_manager_password';  
  
GRANT SELECT ON SkinCare0.Employees TO 'regular_user'@'localhost';
```

```
GRANT SELECT ON SkinCare0.Salaries TO 'regular_user'@'localhost';
```

```
GRANT SELECT, INSERT, UPDATE, DELETE ON SkinCare0.Employees TO  
'hr_manager'@'localhost';
```

```
GRANT SELECT, INSERT, UPDATE, DELETE ON SkinCare0.Salaries TO  
'finance_manager'@'localhost';
```

```
FLUSH PRIVILEGES;
```

```
INSERT INTO Users (UserID, Username, Password, Role)
```

```
VALUES
```

```
(60600, 'admin', SHA2('admin_password', 256), 'Manager'),
```

```
(60601, 'hr_manager', SHA2('hr_pass123', 256), 'HR'),
```

```
(60602, 'sales1', SHA2('sales1_pass', 256), 'Sales'),
```

```
(60603, 'sales2', SHA2('sales2_pass', 256), 'Sales'),
```

```
(60604, 'finance', SHA2('finance_pass123', 256), 'Finance');
```

```
ALTER TABLE Salaries MODIFY BaseSalary BLOB;
```

```
INSERT INTO Salaries(SalaryID, EmployeeID, BaseSalary, Bonus, PaymentDate)
```

```
VALUES ('1',60600, AES_ENCRYPT('5500.00', 'encryption_key'),600.00, '2024-11-27');
```

```
INSERT INTO Salaries(SalaryID, EmployeeID, BaseSalary, Bonus, PaymentDate)
```

```
VALUES ('2',60601, AES_ENCRYPT('8500.00', 'encryption_key'),1200.00, '2024-11-27');
```

```
INSERT INTO Salaries(SalaryID, EmployeeID, BaseSalary, Bonus, PaymentDate)
```

```
VALUES ('3',60603, AES_ENCRYPT('9500.00', 'encryption_key'),2500.00, '2024-11-27');
```

```
INSERT INTO Salaries(SalaryID, EmployeeID, BaseSalary, Bonus, PaymentDate)
```

```
VALUES ('4',60603, AES_ENCRYPT('5500.00', 'encryption_key'),350.00, '2024-11-27');
```

```
INSERT INTO Salaries(SalaryID, EmployeeID, BaseSalary, Bonus, PaymentDate)
```

```
VALUES ('5',60604, AES_ENCRYPT('7500.00', 'encryption_key'),1700.00, '2024-11-27');
```

```
INSERT INTO Salaries(SalaryID, EmployeeID, BaseSalary, Bonus, PaymentDate)
```

```
VALUES ('6',60605, AES_ENCRYPT('6500.00', 'encryption_key'),900.00, '2024-11-27');
```

```
SELECT SalaryID, AES_DECRYPT(BaseSalary, 'encryption_key') AS decrypted_BaseSalary
```

```
FROM Salaries
```

```
WHERE SalaryID = 1;
```

```
SELECT SalaryID, AES_DECRYPT(BaseSalary, 'encryption_key') AS decrypted_BaseSalary
```

```
FROM Salaries
```

```
WHERE SalaryID = 2;
```

```
SELECT SalaryID, AES_DECRYPT(BaseSalary, 'encryption_key') AS decrypted_BaseSalary
```

```
FROM Salaries
```

```
WHERE SalaryID = 3;
```

```
SELECT SalaryID, AES_DECRYPT(BaseSalary, 'encryption_key') AS decrypted_BaseSalary
```

```
FROM Salaries
```

```
WHERE SalaryID = 4;
```

```
SELECT SalaryID, AES_DECRYPT(BaseSalary, 'encryption_key') AS decrypted_BaseSalary
```

```
FROM Salaries
```

```
WHERE SalaryID = 5;
```

```
SELECT SalaryID, AES_DECRYPT(BaseSalary, 'encryption_key') AS decrypted_BaseSalary
```

```
FROM Salaries
```

```
WHERE SalaryID = 6;
```

Code Explanation:

1. Creating and Using the Database

```
CREATE DATABASE SkinCare0;
```

- Creates a new database named `SkinCare0`.

```
USE SkinCare0;
```

- Specifies `SkinCare0` as the active database for subsequent commands.

2. Creating the Employees Table

```
CREATE TABLE Employees (  
    EmployeeID INT PRIMARY KEY,  
    Name VARCHAR(100) NOT NULL,  
    Email VARCHAR(100) NOT NULL UNIQUE,  
    PhoneNumber VARCHAR(15),  
    Position VARCHAR(50) NOT NULL  
);
```

- **EmployeeID:** A unique identifier for each employee, serving as the primary key.
- **Name:** The employee's name; it must not be null.
- **Email:** The employee's email, which must be unique to prevent duplication.
- **PhoneNumber:** The employee's phone number, which can be null.
- **Position:** The employee's job title, which must not be null.

3. Creating the Salaries Table

```
CREATE TABLE Salaries (  
    SalaryID INT,  
    EmployeeID INT NOT NULL,  
    BaseSalary DECIMAL(10, 2) NOT NULL,  
    Bonus DECIMAL(10, 2) DEFAULT 0,  
    PaymentDate DATE NOT NULL,  
    FOREIGN KEY (EmployeeID) REFERENCES Employees(EmployeeID)  
);
```

- **SalaryID:** A unique identifier for the salary record.
- **EmployeeID:** A foreign key referencing the `EmployeeID` in the `Employees` table.
- **BaseSalary:** The employee's base salary, which is mandatory.
- **Bonus:** The additional salary component, with a default value of 0.
- **PaymentDate:** The date the salary is paid, which is mandatory.

4. Creating the Users Table

```
CREATE TABLE Users (  
    UserID INT PRIMARY KEY,  
    Username VARCHAR(50) NOT NULL UNIQUE,  
    Password VARCHAR(256) NOT NULL,  
    Role VARCHAR(20) NOT NULL  
);
```

- **UserID:** A unique identifier for the user, serving as the primary key.
- **Username:** The user's login name, which must be unique.
- **Password:** The user's password, stored in a hashed format for security.
- **Role:** Specifies the user's role in the system (e.g., HR, Finance).

5. Inserting Employee Data

```
INSERT INTO Employees (EmployeeID, Name, Email, PhoneNumber, Position)
VALUES
(60600, 'Sara', 'sara@gmail.com', '0571234506', 'Sales Representative'),
...
(60605, 'Noura', 'noura@gmail.com', '0579988776', 'Marketing Specialist');
```

- Adds records for employees, including their ID, name, email, phone number, and position.

6. Inserting Salary Data

```
INSERT INTO Salaries (SalaryID, EmployeeID, BaseSalary, Bonus, PaymentDate)
VALUES
(1, 60600, 5500.00, 600.00, '2024-11-27'),
...
(6, 60605, 6500.00, 900.00, '2024-11-27');
```

- Adds salary records, linking them to employees using `EmployeeID` and specifying salary, bonus, and payment date.

7. Creating System Users

```
CREATE USER 'regular_user'@'localhost' IDENTIFIED BY
'regular_user_password';
```

- Creates a database user named `regular_user` with the password `regular_user_password`.

```
CREATE USER 'hr_manager'@'localhost' IDENTIFIED BY 'hr_manager_password';
CREATE USER 'finance_manager'@'localhost' IDENTIFIED BY
'finance_manager_password';
```

- Similarly, creates users for HR and Finance managers with their respective passwords.

8. Assigning Permissions

```
GRANT SELECT ON SkinCare0.Employees TO 'regular_user'@'localhost';
```

- Grants `regular_user` permission to only view data (SELECT) in the `Employees` table.


```
GRANT SELECT, INSERT, UPDATE, DELETE ON SkinCare0.Employees TO  
'hr_manager'@'localhost';
```

- Grants `hr_manager` permission to view, insert, update, and delete data in the `Employees` table.

```
GRANT SELECT, INSERT, UPDATE, DELETE ON SkinCare0.Salaries TO  
'finance_manager'@'localhost';
```

- Grants `finance_manager` similar permissions for the `Salaries` table.

```
FLUSH PRIVILEGES;
```

- Ensures all permissions and changes take effect immediately.

9. Storing Users in the Users Table

```
INSERT INTO Users (UserID, Username, Password, Role)  
VALUES  
(60600, 'admin', SHA2('admin_password', 256), 'Manager'),  
...  
(60604, 'finance', SHA2('finance_pass123', 256), 'Finance');
```

- Adds user records with encrypted passwords using the `SHA2` function and specifies their roles.

10. Encrypting Salary Data

```
ALTER TABLE Salaries MODIFY BaseSalary BLOB;
```

- Modifies the `BaseSalary` column to support encrypted data.

```
INSERT INTO Salaries (SalaryID, EmployeeID, BaseSalary, Bonus, PaymentDate)  
VALUES ('1', 60600, AES_ENCRYPT('5500.00', 'encryption_key'), 600.00,  
'2024-11-27');
```

- Encrypts `BaseSalary` using `AES_ENCRYPT` and stores it securely.

11. Retrieving Decrypted Salary Data

```
SELECT SalaryID, AES_DECRYPT(BaseSalary, 'encryption_key') AS  
decrypted_BaseSalary  
FROM Salaries  
WHERE SalaryID = 1;
```

- Retrieves and decrypts the `BaseSalary` for a specific `SalaryID`.

This query is repeated for all salary records, ensuring encrypted data can be securely retrieved.

Output:

Running the code would yield the following results:

Successful creation of the SkinCare0 database and its tables.

```
mysql> CREATE DATABASE SkinCare0;
Query OK, 1 row affected (0.00 sec)

mysql>
mysql> USE SkinCare0;
Database changed
```

1. Employees Table

This table represents basic employee information, including the following columns:

- **EmployeeID:** The employee's unique identifier.
- **Name:** The employee's name.
- **Email:** The employee's email.
- **PhoneNumber:** The employee's phone number.

The data in this table is clear and not encrypted since it doesn't contain sensitive information directly.

```
mysql> SELECT * FROM Employees;
+-----+-----+-----+-----+-----+
| EmployeeID | Name | Email | PhoneNumber | Position |
+-----+-----+-----+-----+-----+
| 60600 | Sara | sara@gmail.com | 0571234506 | Sales Representative |
| 60601 | Layla | layla@gmail.com | 0579876556 | HR Manager |
| 60602 | Mona | mona@gmail.com | 0573456786 | Finance Manager |
| 60603 | Nada | nada@gmail.com | 0571122336 | Sales Representative |
| 60604 | Jana | jana@gmail.com | 0575566776 | Product Manager |
| 60605 | Noura | noura@gmail.com | 0579988776 | Marketing Specialist |
+-----+-----+-----+-----+-----+
6 rows in set (0.00 sec)
```

2. Salaries Table

This table contains information about employee salaries, with some columns encrypted for privacy protection:

- **SalaryID:** The salary record's unique identifier.
- **EmployeeID:** The employee's unique identifier linked to the salary.
- **BaseSalary: Encrypted using AES algorithm.**
- **Bonus:** Additional bonuses.
- **PaymentDate:** The date the salary is paid.

```
mysql> SELECT * FROM Salaries;
```

SalaryID	EmployeeID	BaseSalary	Bonus	PaymentDate
1	60600	0x353530302E3030	600.00	2024-11-27
2	60601	0x383530302E3030	1200.00	2024-11-27
3	60602	0x393530302E3030	2500.00	2024-11-27
4	60603	0x353530302E3030	350.00	2024-11-27
5	60604	0x373530302E3030	1700.00	2024-11-27
6	60605	0x363530302E3030	900.00	2024-11-27
1	60600	0xA90485E5D4456C51537A780B43EC0D02	600.00	2024-11-27
2	60601	0x22B485D214CB4F77467BF602D2C87315	1200.00	2024-11-27
3	60603	0x349E4302958B89435B6BFBD56B87DD1E	2500.00	2024-11-27
4	60603	0xA90485E5D4456C51537A780B43EC0D02	350.00	2024-11-27
5	60604	0xCABD3B97042F4D6528542887F2F7D17C	1700.00	2024-11-27
6	60605	0x51C1E49E15335B69065541375497263F	900.00	2024-11-27

12 rows in set (0.00 sec)

Salary Encryption

The **AES** algorithm was used to encrypt the **BaseSalary** column.

To reveal the original salary (after decryption), the following query was executed:

```
mysql> SELECT SalaryID, AES_DECRYPT(BaseSalary, 'encryption_key') AS decrypted_BaseSalary
-> FROM Salaries
-> WHERE SalaryID = 1;
```

SalaryID	decrypted_BaseSalary
1	NULL
1	0x353530302E3030

2 rows in set (0.00 sec)

```
mysql> SELECT SalaryID, AES_DECRYPT(BaseSalary, 'encryption_key') AS decrypted_BaseSalary
-> FROM Salaries
-> WHERE SalaryID = 2;
```

SalaryID	decrypted_BaseSalary
2	NULL
2	0x383530302E3030

2 rows in set (0.00 sec)

```
mysql> SELECT SalaryID, AES_DECRYPT(BaseSalary, 'encryption_key') AS decrypted_BaseSalary
-> FROM Salaries
-> WHERE SalaryID = 3;
```

SalaryID	decrypted_BaseSalary
3	NULL
3	0x393530302E3030

2 rows in set (0.00 sec)

```
mysql> SELECT SalaryID, AES_DECRYPT(BaseSalary, 'encryption_key') AS decrypted_BaseSalary
-> FROM Salaries
-> WHERE SalaryID = 4;
```

SalaryID	decrypted_BaseSalary
4	NULL
4	0x353530302E3030

2 rows in set (0.00 sec)

```
mysql> SELECT SalaryID, AES_DECRYPT(BaseSalary, 'encryption_key') AS decrypted_BaseSalary
-> FROM Salaries
-> WHERE SalaryID = 5;
```

SalaryID	decrypted_BaseSalary
5	NULL
5	0x373530302E3030

2 rows in set (0.00 sec)

```
mysql> SELECT SalaryID, AES_DECRYPT(BaseSalary, 'encryption_key') AS decrypted_BaseSalary
-> FROM Salaries
-> WHERE SalaryID = 6;
```

SalaryID	decrypted_BaseSalary
6	NULL
6	0x363530302E3030

2 rows in set (0.00 sec)

3. Users Table

This table contains login information for users, with the following columns:

- **UserID:** The user's unique identifier.
- **Username:** The username.
- **Password: Hashed using the SHA algorithm.**
- **Role:** The user's role (e.g., Manager, HR, Sales).

```
mysql> SELECT * FROM Users;
```

UserID	Username	Password	Role
60600	admin	6d4525c2a21f9be1cca9e41f3aa402e0765ee5fcc3e7fea34a169b1730ae386e	Manager
60601	hr_manager	e95fffb48936b1465fc3702bd3d44e8582d73e59f2540fb343eea4addaa2b56e2	HR
60602	sales1	42ff9869475bfdf588fa40a2bee26afe5182925fa53af81abf7aebd3dbf871cb	Sales
60603	sales2	5ba0197b9079a6f87cced66bf3727cb9f72e5951ed49ef872d7f83e56874d65a	Sales
60604	finance	dfc0eb3be61ca0a474bd71c883f9ab930d0752f7255bee0e802da4b062ffc0c6	Finance

```
5 rows in set (0.00 sec)
```

References:

[MySQL AES Functions Documentation](#)

[MySQL SHA2 Function Documentation](#)

[SQL Server HASHBYTES Documentation](#)